SUMMER ANALYTICS 2026 — WEEK 2 HACKATHON

E-Commerce Conversion Prediction Challenge

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1. PROBLEM OVERVIEW:

The objective was to predict whether a user converts on an e-commerce platform. Binary classification problem — Converted = 0 (not converted) or 1 (converted). Evaluated using F1 Score.

MAIN SOLUTION

2. EXPLORATORY DATA ANALYSIS:

Training data: 10,000 samples, 14 features. Target imbalanced — 69.13% non-converted, 30.87% converted. Missing values in Time_On_Site (18.5%), Age (14.8%), Income (9.84%). Two categorical features: Device_Type (Mobile/Desktop/Tablet) and Traffic_Source (5 categories).

3. PREPROCESSING :

Dropped User_ID (unique identifier, no predictive value). Applied one-hot encoding to Device_Type and Traffic_Source using pd.get_dummies(). Used reindex() to align columns across all datasets. Filled missing values using median imputation fit only on training data. Applied StandardScaler fit only on training data to prevent data leakage.

4. FEATURE ENGINEERING:

Created four new features: page_per_time (Pages_Viewed/Time_On_Site), products_per_page (Products_Viewed/Pages_Viewed), Age_Group (age binned into 4 groups), Hot_Lead (binary flag for users with previous purchases AND discount seen).

5. MODEL SELECTION AND RESULTS :

Three models trained and compared on public_test.csv:

Logistic Regression → F1: 0.5438 (SELECTED)

XGBoost → F1: 0.5380

Random Forest → F1: 0.5252

class_weight=balanced used for LR and RF. scale_pos_weight=2.2 for XGBoost. Logistic Regression performed best.

6. FINAL SUBMISSION :

Logistic Regression selected as final model (F1: 0.5438). Predictions

generated for all 3,000 records in private_test.csv and saved as

submission.csv in required format (User_ID, Converted).